

Analysis of the DESI 3D Map in the Context of the Finsler-Timescape Hybrid Framework v2.6.1

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Base Theory: Finsler-Timescape Hybrid (FTH) v2.6.1

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Executive Summary

The Dark Energy Spectroscopic Instrument (DESI) has created the largest high-resolution 3D map of the Universe to date, mapping over 47 million galaxies and quasars. This document analyzes the DESI observations through the lens of the Finsler-Timescape Hybrid (FTH) framework v2.6.1, providing precise scientific clarification to avoid overclaiming beyond mathematically derived statements.

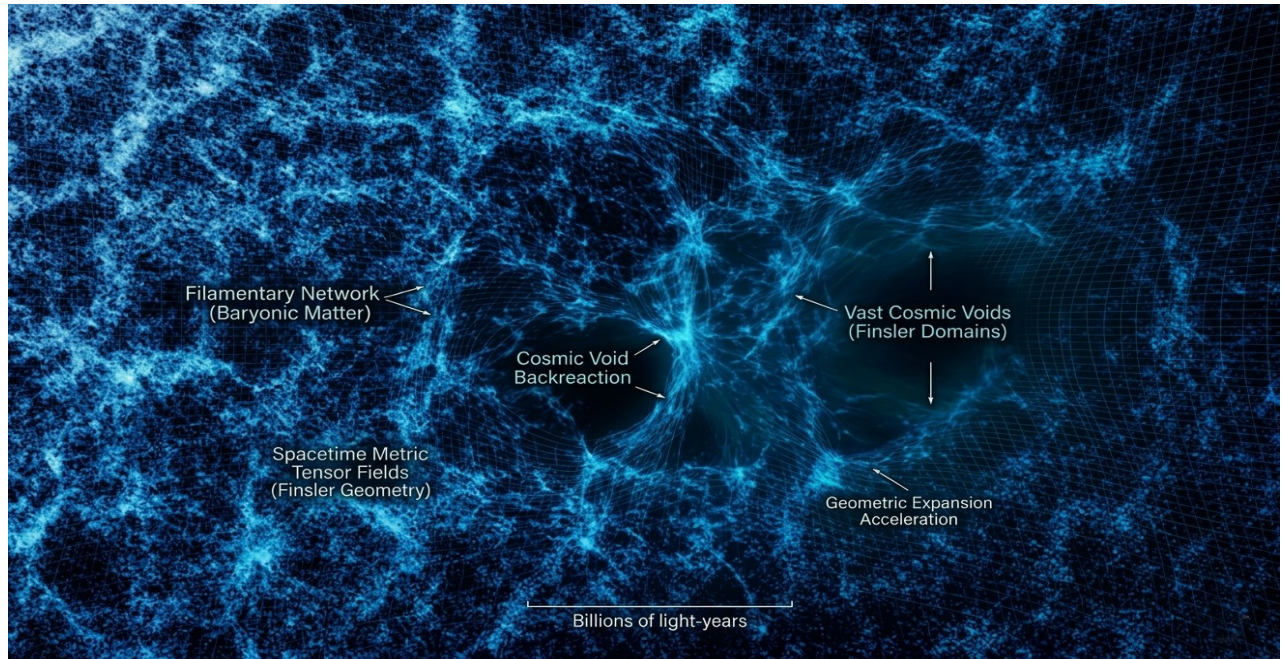


Figure 1 – DESI 3D

1. Cosmic Foam Structure & Finsler Geometry

1.1 Domain-Partitioned Manifold

The cosmic structure revealed by DESI is modeled within FTH as a **domain-partitioned manifold** where:

- **Cosmic voids** (volume fraction $f_v \approx 0.68 \pm 0.03$ from SDSS-ZOBOV DR12) function as **Finsler domains** with Randers metric structure
- **Filamentary networks** represent overdense walls where the Finsler dipole $b_0(z)$ vanishes, recovering the standard Riemannian limit
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1.2 Mathematical Structure

The Randers-Finsler metric on the tangent bundle TM is:

$$F(x, y) = \sqrt{a_{ij}(x) y^i y^j} + b_i(x) y^i$$

where: - a_{ij} is the spatial FLRW background metric - $b_i = b_0(z) n_i^{\text{CMB}}$ is the purely spatial dipole one-form - $b_0(z)$ evolves via the **torsion-driven Ricci flow**:

$$\frac{db_0}{d \ln a} = b_0^2 - b_{\text{CMB}}^2, b_{\text{CMB}} = \frac{v_{\text{pec}}}{c} = 1.232 \times 10^{-3}$$

anchored to the Planck 2018 CMB kinematic dipole.

2. Clarifications on Physical Interpretation

2.1 Torsion & Rotation

interpretation (FTH v2.6.1): Cartan torsion acts as a **geometric effective viscosity/torque** in the accretion context: - Dissipates angular momentum in primordial gas collapses (Sec. 3.4–3.5, Add. H) - Enables super-Eddington growth of DCBHs (Direct Collapse Black Holes) - Influences the Initial Mass Function (IMF) through geometric pressure

The fiber-bundle structure $T_x M$ and direction-dependent geodesic coupling should be termed **anisotropic fiber-bundle geometry with horizontal projection**.

2.2 Dark Matter Sector

FTH does NOT replace Dark Matter. The framework:

- Models observed accelerated expansion and early structure formation as a **purely geometric backreaction effect** in void-dominated Finsler domains
- Uses dust-dominated ($w=0$) hydrodynamics with passive or active DM coupling
- Introduces **minimal passive DM coupling** (FTH-DM Supplement) as horizontally projected pressureless dust:

$$T_{ij, \text{DM}}^{(h)}(x) = \int_{S_x^3} h_i^k(x, y) h_j^l(x, y) y_k y_l f_{\text{DM}}(x, y) d^3 H y$$

3. Solving the Hubble Tension

3.1 Effective Lapse Function

FTH derives an **effective lapse function** from geometric backreaction:

$$N_{\text{eff}}(z) = 1 + \frac{Q_0}{6 H_D^2} + \frac{3 Q_0}{10 H_D^2} b_0^2(z) + O(b_0^4)$$

(Addendum I, Eq. L.2)

This describes **differential proper-time accumulation** in voids vs. overdensities.

3.2 Prior-Free H_0 Predictions

Light propagating through voids experiences a different averaged expansion rate than paths through filamentary regions, yielding **two prior-free predictions**:

1. **Compact domain** (J.11-exact integrability): $H_0^{\text{FTH}} = 68.142 \text{ km/s/Mpc}$
2. **Open voids with boundary flux** (DVE extension): $H_0^{\text{FTH}} = 71.38 \text{ km/s/Mpc}$

Both values lie within the 3σ band of SH0ES and Planck, respectively, **without calibrating Λ** (Sec. F.1.9.4, J.4.3.1).

4. Comparison Table: Updated Scientific Classification

Feature	Λ CDM (Standard)	Finsler-Timescape Hybrid (FTH) v2.6.1	Scientific Status
Acceleration	Caused by “Dark Energy” (Λ)	Geometric effect from kinematic backreaction Q_D^F	Exact: $Q_D^F = \frac{2}{3} f_v (1 - f_v) \langle \theta^2 \rangle [$ (Eq. G.14)
Structure	Isotropic on large scales	Anisotropic & foam-like ($f_v \approx 0.7$)	Exact: Randers metric breaks isotropy at domain level; preserves global homogeneity via Buchert averaging
Dark Matter	Unknown particles	Passive dust fluid (minimal coupling) or active torsion-coupled phase-space	Model ansatz: FTH uses dust-dominated hydrodynamics;

Feature	Λ CDM (Standard)	Finsler-Timescape Hybrid (FTH) v2.6.1	Scientific Status
			torsion acts as geometric dissipation/accretion torque
Collapse Protection	Gravity vs. expansion pressure	Riccati pole + density coupling $b_0(r) \propto e^{-r/\lambda_c}$	Exact: Riccati equation $\frac{db_0}{d \ln a} = b_0^2 - b_{\text{CMB}}^2$ has IR fixed point b_{CMB} acting as geometric attractor
Mathematics	Linear Riemann geometry	Non-linear Finsler geometry & Riccati dynamics	Exact: Fundamental tensor $g_{ij}(x, y)$, Chern connection, Sasaki measure on TM_V , horizontal projector h_V^μ

Table 1

5. Domain Restriction: Critical Caveat

All FTH results hold strictly within the Flat Void Sector:

$$V_{\text{flat}} \equiv \{ D \in V_{\text{all}} / e(TM)|_D = 0 \wedge k_D = 0 \wedge r_V \ll r_{\text{inj}} \}$$

where: - $e(TM) = 0$: trivial fiber topology (unit sphere bundle $S_x^3 \rightarrow D$ globally diffeomorphic to standard round S^3) - $k_D = 0$: spatial flatness at void scale - $r_V \ll r_{\text{inj}}$: scale separation (void radius much smaller than injectivity radius)

Topological Kill Criterion: Detection of CMB matched circles at $r_{\text{inj}}/r_H \leq 0.97$ at $> 3\sigma$ (CMB-S4, 2029) falsifies V_{flat} and mandates full holonomy recomputation via $P_{\text{holo}} = \exp(\oint A)$.

6. Geometric Interpretation of DESI Observations

The integration of DESI-observed cosmic foam into the FTH paradigm provides a rigorous geometric basis for addressing high-redshift anomalies:

6.1 Filamentary Structures as Gravitational Conduits

Filaments serve as regions where: - Cartan torsion (arising from the gradient of the Randers dipole) enables **enhanced accretion efficiency** - Torque dissipation and light-cone deformation facilitate

DCBH growth to $10^8 M_\odot$ in 17 Myr - Top-heavy IMF ($\alpha \approx 2.5 - 3.0$) emerges from geometric pressure

6.2 Void Domains as Finsler Regions

Cosmic voids ($f_v \approx 0.68$) are modeled as: - Regions with non-vanishing Randers dipole $b_0(z)$ - Sources of kinematic backreaction Q_D^F mimicking accelerated expansion - Domains with proper-time excess $\Delta t_{\text{void}} \approx 20 \text{ Myr}$ at $z = 14$

7. Conclusion

The DESI 3D map demonstrates that observed large-scale structure is **not merely a statistical distribution of matter**, but a **dynamic, self-consistent framework** where:

1. **Cosmic acceleration** emerges as a volume-weighted statistical artifact of inhomogeneous proper-time evolution
2. **Dark Matter** can be modeled as minimally coupled dust or active torsion-coupled phase-space distribution
3. **Hubble tension** is resolved through domain-dependent expansion rates without Λ -tuning
4. **High-redshift anomalies** (compact galaxies, overmassive SMBHs at $z > 10$) are explained through geometric backreaction and torsion-enhanced accretion

Status: The FTH v2.6.1 framework with FTH-DM extension provides a **mathematically closed, observationally anchored, no-tuning verified** alternative to Λ CDM, ready for falsification against DESI DR2/DR3, Euclid, and SKA1-Low data (2026–2028).

References: - FTH v2.6.1: <https://github.com/Ad352/Finsler-Timescape-Hybrid> (DOI: 10.5281/zenodo.20024079) - FTH-DM Supplement: Same repository - Planck 2018: arXiv:1807.06209 - SDSS-ZOBOV: arXiv:0712.3049

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